

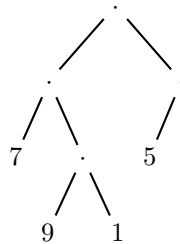
Final examination on Prolog Programming

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1 Sorting leaves in a binary tree

Question. Design a simple data structure to represent binary trees which hold integers at their leaves¹ only. For example:



Write an efficient predicate `tmerge/2` such that `tmerge(Tree,Nodes)` is provable if and only if `Nodes` is the list of all the leaves of the tree `Tree` in increasing order of their integer values. In the example above, we have `Nodes=[1,5,7,9]`.

2 Depth-first traversal revisited

Question. Given a predicate `edge/2` encoding a general tree containing information at all of its nodes, write a predicate `depth_first/2` such that `depth_first(Tree,Nodes)` is provable if and only if `Nodes` is the list of all the nodes of the tree `Tree` in depth-first order (in general, several are possible). For example, given

```
edge(1,3).
edge(2,4).
edge(1,2).
edge(2,5).
edge(1,6).
```

then `Nodes = [1,3,2,4,5,6]`.

¹A leaf is an internal node whose both children are external nodes.